

Multiplying and Dividing in Scientific NotationQuestion Bank - Additional 4.02 **ANSWER KEY***Show your work. Circle your final answer.***For questions 1 through 6, multiply or divide. Write each answer in scientific notation.**

1. $(3 \times 10^2)(3 \times 10^6)$

9×10^8

2. $\frac{6.3 \times 10^7}{2.1 \times 10^3}$

3×10^4

3. $(4 \times 10^5)(8 \times 10^6)$

$32 \times 10^{11} = 3.2 \times 10^{12}$

4. $\frac{2.4 \times 10^4}{8 \times 10^7}$

$0.3 \times 10^{-3} = 3 \times 10^{-4}$

5. $(2.5 \times 10^{-5})(6 \times 10^{-4})$

$15 \times 10^{-9} = 1.5 \times 10^{-8}$

6. $\frac{9.9 \times 10^{-3}}{3.3 \times 10^{-8}}$

3×10^5

$-3 - (-8) = 5$

Real quantities.

7. The Amazon River discharges about 5.5×10^7 gallons of water every second.

a. About how many gallons does it discharge in one hour, which is 3.6×10^3 seconds?

about 1.98×10^{11} gallons

$5.5 \cdot 3.6 = 19.8$, so the product is 19.8×10^{10} before it is finished.

b. A second river discharges about 1.1×10^7 gallons per second. How many times larger is the Amazon's rate?

5 times larger

8. A factory produces about 6.4×10^5 bottles per day.

a. How many bottles does it produce in 3.5×10^2 days?

$$2.24 \times 10^8 \text{ bottles}$$

$$6.4 \cdot 3.5 = 22.4 \text{ .}$$

b. A second factory produced 7.2×10^8 bottles in 1.2×10^3 days. How many bottles per day is that?

$$6 \times 10^5 \text{ bottles per day}$$

Working backwards.

9. Find the missing number in each equation.

a. $(4 \times 10^n)(2 \times 10^3) = 8 \times 10^{10}$

$$n = 7$$

$$n + 3 = 10 \text{ .}$$

b. $\frac{7.2 \times 10^6}{n \times 10^2} = 2.4 \times 10^4$

$$n = 3$$

$$7.2 \div n = 2.4 \text{ .}$$

c. $(2.5 \times 10^3)(8 \times 10^5) = \underline{\hspace{2cm}}$

$$2 \times 10^9$$

The coefficients multiply to exactly 20.

Reading someone else's work.

10. Dana divides $\frac{4.8 \times 10^{11}}{1.2 \times 10^4}$ and writes 3.6×10^7 . Name her mistake and give the correct quotient.

She subtracted the coefficients instead of dividing them. The power of ten is right. The correct quotient is 4×10^7 .

$$4.8 \div 1.2 = 3.6 \text{ , which is where her coefficient came from; } 4.8 \div 1.2 = 4 \text{ .}$$

11. Which is the value of $\frac{9.6 \times 10^{-3}}{3.2 \times 10^5}$?

● A. 3×10^{-8}

○ B. 3×10^8

○ C. 6.4×10^{-8}

○ D. 3×10^2

A

12. Select ALL of the expressions whose value is 8×10^6 .

☒ **A.** $(1.6 \times 10^3)(5 \times 10^3)$

☒ **B.** $\frac{1.6 \times 10^{10}}{2 \times 10^3}$

☐ **C.** $(4 \times 10^2)(2 \times 10^8)$

☒ **D.** $\frac{4 \times 10^8}{5 \times 10^1}$

☒ **E.** $(4 \times 10^2)(2 \times 10^4)$

☐ **F.** $\frac{1.6 \times 10^{10}}{2 \times 10^4}$

A, B, D and E

B and D each give 0.8×10^7 , which is 8×10^6 once it is finished.